INNOPOLIS UNIVERSITY

SNE MASTER TRACK

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COURSE: [F25] Networks Engineering

NE Lab 4 Report

On

OSPF

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Task 1

1. In the GNS3 project, select and install a virtual routing solution that you would like to use: Mikrotik (recommended), Pfsense, vyos and so on.

Answer:

I have installed Mikrotik router for installing virtual routing solution for this GNS3 project.

2. Prepare a simple network consisting of at least 3 routers, each one of them has a different subnet, and they should be able to reach each other (for example by a a) b) switch/router in the middle or a bus topology). Do not write a static routes between different networks.

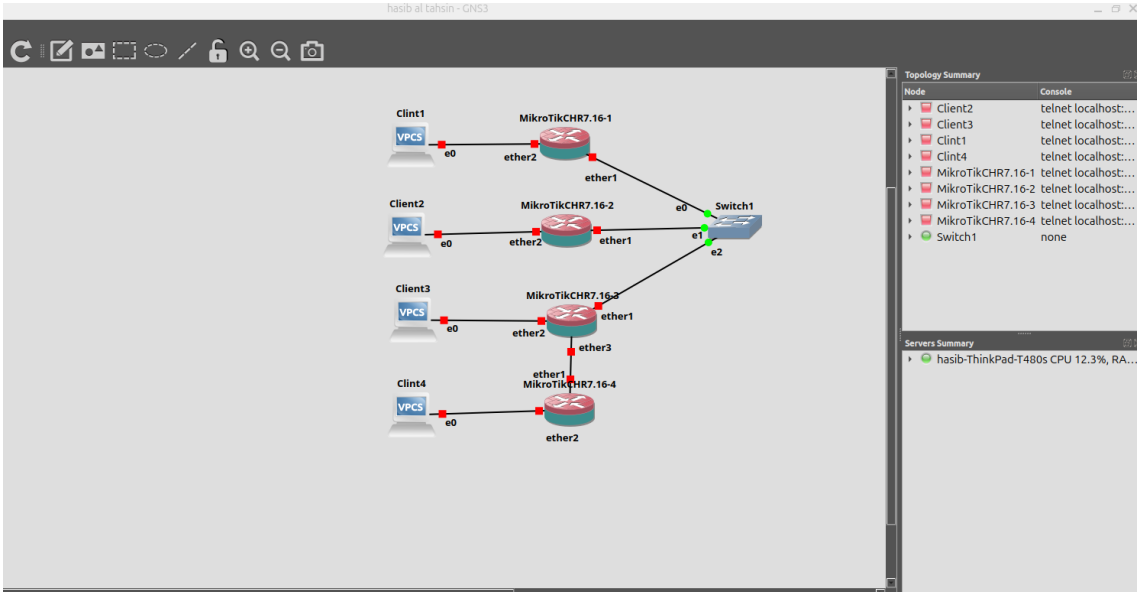
Answer:

In this lab, I have created a complete multi-router network using 4 MikroTik RouterOS routers. *Router1, Router2, Router3* are connected to a backbone network (*10.0.0.0/24*) that ensures Layer-3 connectivity without any static routes [1]. *Router3* and *Router4* are connected to a separate Point-to-Point network (*10.0.1.0/30*), which follows the criteria of real-world router interconnection [5]. Clients are connected using VPCS [4].

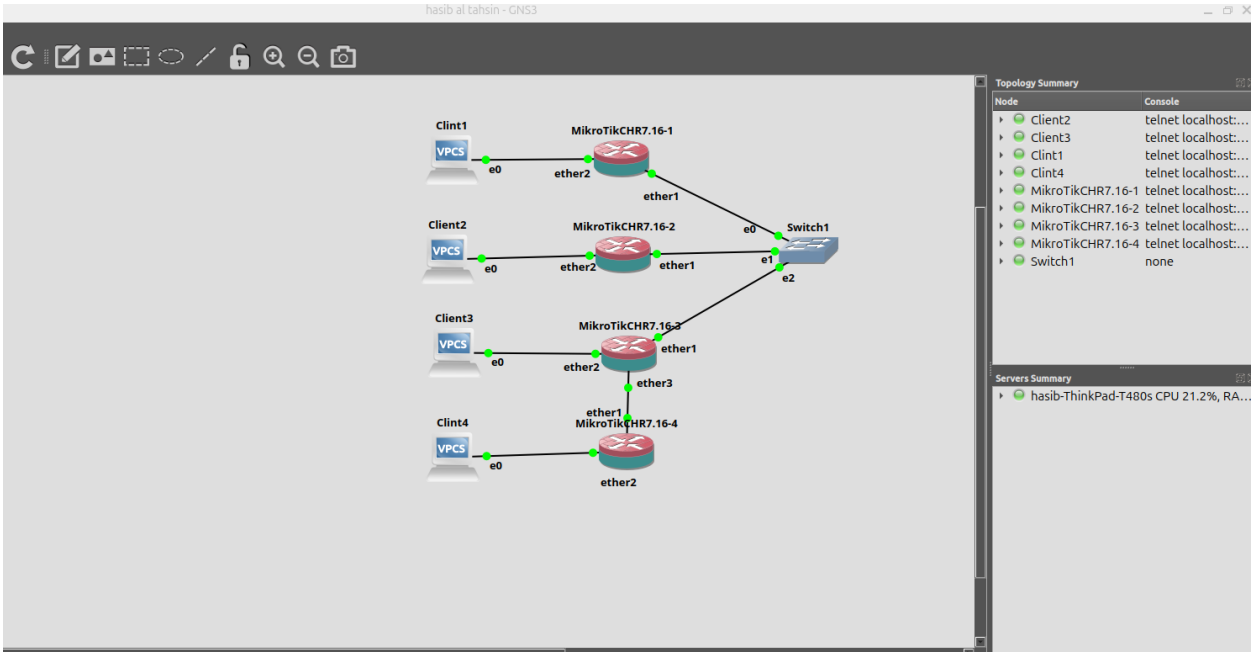
Topology

Each router has a separate LAN client connected to it. This type of design is used as the standard topology for GNS3 network simulation [2].

At off situation:



When it was turn on:



Device:

- Router1
- Router2
- Router3
- Router4
- Switch
- Client1
- Client2
- Client3
- Client4 (VPCS)

IP Plan

This backbone network has created a shared L3 domain similar to OSPF Area 0 [3].

Link	Device	Interface	IP / Subnet
Backbone Network	*Router1*	*ether1*	*10.0.0.1/24*
(With Switch	*Router2*	*ether1*	*10.0.0.2/24*
R1-R2-R3)	*Router3*	*ether1*	*10.0.0.3/24*
R3–R4 Point-to-Point	*Router3*	*ether3*	*10.0.1.1/30*
	Router4	*ether1*	*10.0.1.2/30*
LAN1	*Router1*	*ether2*	*192.168.1.1/24*
LAN2	*Router2*	*ether2*	*192.168.2.1/24*
LAN3	*Router3*	*ether2*	*192.168.3.1/24*
LAN4	*Router4*	*ether2*	*192.168.4.1/24*

Client IP (VPCS)

Client	IP / Mask	Gateway
Client1	*192.168.1.10/24*	*192.168.1.1*
Client2	*192.168.2.10/24*	*192.168.2.1*
Client3	*192.168.3.10/24*	*192.168.3.1*
Client4	*192.168.4.10/24*	*192.168.4.1*

Setting Hostname of Router1

```
[admin@MikroTik] > /system identity set name=Router1
```

This command changes the router name (hostname). Earlier router name was default *MikroTik1*. After running this command, the name of the router becomes **Router1**.

Setting IP Address

Backbone link *Router1-Switch-Router2,3*

```
[admin@Router1] > /ip address add address=10.0.0.1/24 interface=ether1 comment="Backbone LAN"
```

This command sets the *10.0.0.1/24* IP on *Router1*'s ether1 interface. This subnet (*10.0.0.0/24*) is the OSPF Backbone Network (Area 0). *Router1, Router2* and *Router3* are connected to each other in this backbone. Commenting "*Backbone LAN*" makes it easier to indicate in the interface table that it is a backbone link.

Local LAN1 Router1-Client1

```
[admin@Router1] > /ip address add address=192.168.1.1/24 interface=ether2 comment="LAN1"
```

This command adds *192.168.1.1/24* IP to *Router1*'s ether2 interface. This is *Router1*'s Local LAN 1, where Client1 (PC1) is connected. The IP served as the gateway for *Client1* (*192.168.1.1*). Comment "*LAN1*" indicates that this is the local network interface.

Connection Test

Pinging *Router1*'s own Backbone interface

```
admin@Router1] > ping 10.0.0.1
 SEQ HOST                                     SIZE TTL TIME       STATUS
   0 10.0.0.1                                   56  64 112us
   1 10.0.0.1                                   56  64 207us
   2 10.0.0.1                                   56  64 172us
   3 10.0.0.1                                   56  64 124us
   4 10.0.0.1                                   56  64 120us
   5 10.0.0.1                                   56  64 119us
   sent=6 received=6 packet-loss=0% min-rtt=112us avg-rtt=142us
  max-rtt=207us
```

The reply was received successfully from *10.0.0.1* and no packet-loss found.

Ping Router1's own LAN IP

```
[admin@Router1] > ping 192.168.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.1.1                                56  64 109us
    1 192.168.1.1                                56  64 121us
    2 192.168.1.1                                56  64 180us
    sent=3 received=3 packet-loss=0% min-rtt=109us avg-rtt=136us
   max-rtt=180us
```

The reply was received successfully from *192.168.1.1* and no packet-loss found.

Router2 Config

Setting Hostname of Router2

```
[admin@MikroTik] > /system identity set name=Router2
```

This command changes the hostname of *Mikrotik2* which is set to *Router2*. It helps to distinguish *Router2* in OSPF neighbor table, routing debug and topology diagram.

Setting IP Address

Backbone (from Switch, R1, R3)

```
[admin@Router2] > /ip address add address=10.0.0.2/24 interface=ether1 comment="Backbone LAN"
```

10.0.0.2/24 IP is assigned to *Router2*'s ether1 interface. This subnet *10.0.0.0/24* is OSPF's Area 0 Backbone Network. Through this backbone *Router1, Router2* and *Router3* can detect each other as neighbors. Looking at the "*Backbone LAN*" comment config makes it easy to understand that this is the core OSPF link [1].

LAN2 Router2-Client2

```
[admin@Router2] > /ip address add address=192.168.2.1/24 interface=ether2 comment="LAN2"
```

Router2 local LAN IP is set to *192.168.2.1/24* on ether2 interface. *Client2 (PC2)* used this IP as the gateway. This subnet is used between *Router2* and *Client2*.

Direct Link Ping

At Own Backbone IP

```
[admin@Router2] > ping 10.0.0.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.0.2                                   56  64 128us
    1 10.0.0.2                                   56  64 120us
    2 10.0.0.2                                   56  64 118us
    3 10.0.0.2                                   56  64 221us
    4 10.0.0.2                                   56  64 175us
    sent=5 received=5 packet-loss=0% min-rtt=118us avg-rtt=152us max-rtt=221us
```

Direct ping same net from Router2 to Router1

```
[admin@Router2] > /ping 10.0.0.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.0.1                                   56  64 2ms550us
    1 10.0.0.1                                   56  64 2ms
    2 10.0.0.1                                   56  64 2ms3us
    3 10.0.0.1                                   56  64 2ms257us
    4 10.0.0.1                                   56  64 2ms155us
    5 10.0.0.1                                   56  64 905us
    6 10.0.0.1                                   56  64 1ms813us
    sent=7 received=7 packet-loss=0% min-rtt=905us avg-rtt=1ms954us max-rtt=2ms550us
```

The reply was received successfully from 10.0.0.1 and no packet-loss found. Router2 to Router1 is pinged on the same subnet (10.0.0.0/24). This ping is used to verify that the backbone link is working properly. Because both reside in the same Layer 3 subnet [1].

Router3 Config (Link to Backbone + R4 + LAN3)

Setting Hostname of Router3

```
[admin@MikroTik] > /system identity set name=Router3
```

This command sets the router's hostname to *Router3*. *Router3* is the ABR (Area Border Router) of this lab, because it connects Area0 (Backbone) and Area1 [1].

Setting IP Address

With Backbone LAN Switch

```
[admin@Router3] > /ip address add address=10.0.0.3/24 interface=ether1 comment="Backbone LAN"
```

Added IP 10.0.0.3/24 to *Router3*'s ether1 interface. This subnet (10.0.0.0/24) was OSPF's Area 0 Backbone network. Here *Router1, Router2* and *Router3* were all connected to the same layer-3 network. OSPF adjacency and LSDB sync run through the backbone [1].

LAN3

```
[admin@Router3] > /ip address add address=192.168.3.1/24 interface=ether2 comment="LAN3"
```

Router3's ether2 interface has the Local LAN IP set to *192.168.3.1/24*. Since there are clients on this subnet, their gateway is *Router3* (*192.168.3.1*). This network is advertised in OSPF Area1 (resulting in a summary route to Area0) [5].

Point-to-point link to Router4

```
[admin@Router3] > /ip address add address=10.0.1.1/30 interface=ether3 comment="Link to Router4"
```

A point-to-point link (*10.0.1.0/30*) is set between *Router3* and *Router4*. *Router3*'s IP is *10.0.1.1/30, Router4's* IP is *10.0.1.2/30* and this link is part of Area1 (non-backbone area) [5]. *Router3* sends routing information from Area1 to Area0 as an ABR [5].

Direct Link Ping

on own Backbone IP

```
[admin@Router3] > ping 10.0.0.3
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.0.3                                   56  64 100us
    1 10.0.0.3                                   56  64 116us
    2 10.0.0.3                                   56  64 116us
    3 10.0.0.3                                   56  64 119us
    4 10.0.0.3                                   56  64 118us
    5 10.0.0.3                                   56  64 118us
    sent=6 received=6 packet-loss=0% min-rtt=100us avg-rtt=114us max-rtt=119us
```

The ping got the reply and checked that the backbone interface (ether1) of *Router3* is up properly [1].

Without Static Route

```
[admin@Router3] > /ping 10.0.0.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.0.1                                   56  64 2ms213us
    1 10.0.0.1                                   56  64 2ms83us
    2 10.0.0.1                                   56  64 2ms348us
    3 10.0.0.1                                   56  64 2ms244us
    4 10.0.0.1                                   56  64 2ms44us
    5 10.0.0.1                                   56  64 1ms964us
    sent=6 received=6 packet-loss=0% min-rtt=1ms964us avg-rtt=2ms149us max-rtt=2ms348us
```

```
[admin@Router3] > /ping 10.0.0.2
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.0.2                                   56  64 4ms96us
    1 10.0.0.2                                   56  64 2ms103us
    2 10.0.0.2                                   56  64 1ms988us
    3 10.0.0.2                                   56  64 2ms316us
    sent=4 received=4 packet-loss=0% min-rtt=1ms988us avg-rtt=2ms625us max-rtt=4ms96us
```

Pinged *Router1 (10.0.0.1)* and *Router2 (10.0.0.2)* from *Router3* and got replied. Since all three were on the same subnet (*10.0.0.0/24*), no static route was needed. This test confirmed that *R1, R2* and *R3* backbone connectivity was OK [1].

Router4 Config

Setting Hostname

```
[admin@MikroTik] > /system identity set name=Router4
```

This command sets the name of Router4.

IP Address

Point-to-point link to Router3

```
[admin@Router4] > /ip address add address=10.0.1.2/30 interface=ether1 comment="Link to Router3"
```

This command adds IP *10.0.1.2/30* to the ether1 interface of *Router4*. This link creates a point-to-point connection between *Router3* and *Router4* (*10.0.1.0/30* network) [1].

LAN4

```
[admin@Router4] > /ip address add address=192.168.4.1/24 interface=ether2 comment="LAN4"
```

Router4's ether2 interface has Local LAN IP set to *192.168.4.1/24* This network acts as a gateway for *Client4* [3].

Direct Link Ping

On own P2P IP

```
[admin@Router4] > ping 10.0.1.2
 SEQ HOST                                     SIZE TTL TIME       STATUS
   0 10.0.1.2                                   56  64 181us
   1 10.0.1.2                                   56  64 115us
   2 10.0.1.2                                   56  64 120us
   3 10.0.1.2                                   56  64 258us
   sent=4 received=4 packet-loss=0% min-rtt=115us avg-rtt=168us max-rtt=258us
```

This command pinged *Router4* on its own ether1 interface IP (10.0.1.2) and got a reply.

P2P IP ping of R4 to R3

```
[admin@Router4] > ping 10.0.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.0.1.1                                   56  64 2ms992us
    1 10.0.1.1                                   56  64 1ms751us
    2 10.0.1.1                                   56  64 1ms816us
    3 10.0.1.1                                   56  64 1ms707us
    4 10.0.1.1                                   56  64 1ms650us
    sent=5 received=5 packet-loss=0% min-rtt=1ms650us avg-rtt=1ms983us max-rtt=2ms992us
```

Router4 to *Router3*'s P2P IP (10.0.1.1) is pinged. Since the two are on the same 10.0.1.0/30 subnet, this ping is successful [3].

Clients (VPCS) Config

Client1 (with Router1)

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

Here *Client1* is set to IP *192.168.1.10* and gateway *192.168.1.1* (LAN interface of *Router1*) [4].

Testing

Gateway R1

```
PC1> /ping 192.168.1.10

192.168.1.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=5 ttl=64 time=0.001 ms
```

Router1 Gateway was pinged from *Client1* and received a reply.

Reply from own IP

```
PC1> /ping 192.168.1.10

192.168.1.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.10 icmp_seq=5 ttl=64 time=0.001 ms
```

Check your IP by pinging to see if the VPCS network stack is running properly [3].

Client2 (With Router2)

```
PC2> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC2 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

Client2's IP address is *192.168.2.10* and gateway is *Router2*'s LAN2 interface *192.168.2.1.* This adds *Client2* to *Router2*'s network (*192.168.2.0/24*) [4].

Ping Test:

```
PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=64 time=1.246 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=64 time=1.361 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=64 time=1.430 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=64 time=1.489 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=64 time=1.460 ms
^C
```

Layer 2 and Layer 3 connections are verified by pinging gateway *192.168.2.1 (Router2)* from *Client2* [3].

Client3 (With Router3)

```
Executing the startup file

PC3> ip 192.168.3.10/24 192.168.3.1
Checking for duplicate address...
PC3 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1
```

With this, *Client3* is set to IP *192.168.3.10* and *Router3*'s LAN interface *192.168.3.1* is used as gateway [3].

Ping:

```
PC3> ping 192.168.3.1

84 bytes from 192.168.3.1 icmp_seq=1 ttl=64 time=1.171 ms
84 bytes from 192.168.3.1 icmp_seq=2 ttl=64 time=1.343 ms
84 bytes from 192.168.3.1 icmp_seq=3 ttl=64 time=1.344 ms
84 bytes from 192.168.3.1 icmp_seq=4 ttl=64 time=1.616 ms
^C
```

By pinging *Router3* gateway IP from *Client3*, connectivity is checked [1].

Client4 (With Router4)

```
PC4> ip 192.168.4.10/24 192.168.4.1
Checking for duplicate address...
PC4 : 192.168.4.10 255.255.255.0 gateway 192.168.4.1
```

Client4 is set to *192.168.4.10* IP + gateway *192.168.4.1 Router4*'s LAN4 interface [4].

Ping:

```
PC4> ping 192.168.4.1

84 bytes from 192.168.4.1 icmp_seq=1 ttl=64 time=1.315 ms
84 bytes from 192.168.4.1 icmp_seq=2 ttl=64 time=1.314 ms
84 bytes from 192.168.4.1 icmp_seq=3 ttl=64 time=1.338 ms
84 bytes from 192.168.4.1 icmp_seq=4 ttl=64 time=1.569 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=64 time=1.091 ms
^C
```

Local connectivity is verified by pinging *Router4*'s gateway IP from *Client4* and getting a reply [3].

TASK 2 - OSPF Learning & Configuring

1. Deploy OSPF in your chosen network topology

Answer:

This network consists of a total of 4 MikroTik RouterOS routers, which have OSPF enabled as of RouterOS v7 [1]:

- Router1 - Area 0
- Router2 - Area 0
- Router3 - Area 0 + Area 1 (ABR)
- Router4 - Area 1

OSPF configuration requires:

- Adding an OSPF instance
- Adding an area
- Advertising the network via interface-template (RouterOS v7 standard)
- Enabling authentication (according to security best practices) [1]

Router 1 - OSPF (Area 0)

```
[admin@Router1] > /routing ospf instance add name=default router-id=1.1.1.1
```

Here, an OSPF instance has been created on Router1, named default, and its router-id is set to *1.1.1.1*. The OSPF router-id is the unique identity of the router, which is used in OSPF database, LSA, DR/BDR election [3].

```
[admin@Router1] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

This creates an area called area0 in OSPF with area-id = *0.0.0.0*.

Area 0 is the Backbone Area of OSPF, where all non-backbone areas are ultimately connected. Router1 is part of this area0 [3].

```
[admin@Router1] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Here *10.0.0.0/24* (Backbone LAN) is advertised in OSPF within area0.

Router1, Router2, Router3 are together in this network, so they detect each other as OSPF neighbors. OSPF authentication is enabled with *auth=simple auth-key=labpass* so all routers are given the same password [1], [2].

```
[admin@Router1] > /routing ospf interface-template add networks=192.168.1.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Router1's local LAN1 network *192.168.1.0/24* is also included in area0.

This allows other routers to learn about the *192.168.1.0/24* subnet via OSPF and get a route to Client1's network [3].

Router 2 - OSPF (Area 0)

```
[admin@Router2] > /routing ospf instance add name=default router-id=2.2.2.2
```

An OSPF instance is also created on Router2, with router-id = *2.2.2.2*. This is different from *Router1*'s router-id, so that there is no conflict in the OSPF database [3].

```
[admin@Router2] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

Router2 is also joined to backbone area0, just like *Router1*. This makes *Router2* part of the backbone and forms an OSPF adjacency with R1 and R3 [3].

```
[admin@Router2] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpass
```

Backbone network 10.0.0.0/24 is enabled in OSPF of Router2.

It appears as a neighbor with *Router2, Router1* and *Router3*. Here too, Authentication key labpass [3].

```
[admin@Router1] > /routing ospf interface-template add networks=192.168.1.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Router2's local LAN2 (*192.168.2.0/24*) is advertised in area0 [1].

Router 3 - OSPF (Area 0 + Area 1)

Router3 is ABR

```
[admin@Router3] > /routing ospf instance add name=default router-id=3.3.3.3
```

An OSPF instance is created on *Router3* and the router-id is set to *3.3.3.3*. This router-id is used for LSAs and inter-area routing when acting as an ABR [3].

```
[admin@Router2] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

Router3 is made part of backbone area0 so that it can participate in the backbone by becoming an OSPF neighbor with *Router1* and *Router2* [3].

```
[admin@Router3] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
```

Here a new OSPF area has been created - area1 (area-id *0.0.0.1*). This is the non-backbone area where *Router3* and *Router4* operate. *Router3* now stands on the boundary of area0 and area1 together, it is called ABR (Area Border Router).

Area 0

```
[admin@Router3] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpass
```

Backbone link 10.0.0.0/24 is enabled in area0 for *Router3*. It creates OSPF adjacency with *Router1* and *Router2* and syncs the backbone LSDB.

Area 1

```
[admin@Router3] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1 auth=simple auth-key=labpass
```

Here, *Router3* and *Router4* have a point-to-point link (*10.0.1.0/30*) added to area1. This creates an OSPF neighbor relationship between R3 and R4, but this link is not a backbone, as it is part of Area1 [3].

```
[admin@Router3] > /routing ospf interface-template add networks=192.168.3.0/24 area=area1 auth=simple auth-key=labpass
```

Here, *Router3*'s LAN3 subnet (*192.168.3.0/24*) is advertised in area1.

The route of this subnet is propagated to both area1 and area0 via OSPF due to the presence of ABR [3].

Router 4 – OSPF (Area 1)

```
[admin@Router4] > /routing ospf instance add name=default router-id=4.4.4.4
```

An OSPF instance has been created on *Router4* with router-id *4.4.4.4*.

```
[admin@Router4] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
```

Router4 is only part of area1. It is not directly connected to backbone area0, but participates in Area1 through *Router3* [3].

```
[admin@Router4] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1 auth=simple auth-key=labpass
```

Router4's P2P link (*10.0.1.0/30*) is enabled in OSPF Area1 with *Router3*.

```
[admin@Router4] > /routing ospf interface-template add networks=192.168.4.0/24 area=area1 auth=simple auth-key=labpass
```

Router4's local *LAN4* subnet (*192.168.4.0/24*) is advertised in OSPF Area1. As a result, other routers R1, R2, R3 can learn about this network through OSPF and get a route to *Client4* [3].

2. Which interface you will select as the OSPF router ID and why?

Answer:

The OSPF Router-ID is a 32-bit number that looks like an IPv4 address, but it is not a real IP address. The Router-ID is the identity used to uniquely identify a router within OSPF. The Router-ID should never change because changing the Router-ID restarts the entire OSPF topology, LSA database, neighbor relationships, and DR/BDR election, which can make the network unstable. For this reason, the IP address of a physical interface is not selected as the Router-ID, because if the interface goes down or the IP changes, the Router-ID is forced to change. Therefore, the best practice in OSPF is to use a loopback interface or manually assigned logical router-id, which do

not change. In this lab, a different manual Router-ID is used for each router (*1.1.1.1, 2.2.2.2, 3.3.3.3*, and *4.4.4.4*), which is fully consistent with best practice. This results in stable OSPF databases, reliable DR/BDR elections, and clean and consistent network designs [3].

3. What is the difference between advertising all the networks VS manual advertising (per interface or per subnet)? Which one is better?

Answer:

If advertise all networks is used in OSPF, the router automatically scans all its interfaces and sends every connected subnet to OSPF. As a result, many unwanted subnets such as client LAN, test VLAN, management interface, DHCP-only port can enter OSPF and spread unnecessarily throughout the network. This type of configuration increases security risks, because internal client networks and sensitive interfaces are unintentionally advertised. Moreover, the design also becomes dirty and troubleshooting becomes difficult in large networks [3]. In contrast, manual per-interface OSPF advertising only adds the necessary subnets to OSPF. As a result, the design remains clean and controlled, security is maintained, and unnecessary entries are not accumulated in the OSPF database. This is an industry-standard best practice and this method was followed in Lab Task 2 as well. In this lab, only those subnets that needed to be sent to OSPF were advertised using manual interface-templates [1], [3]. Therefore, the manual method is the correct, safe solution.

4. If you have a static route in a router, how can you let your OSPF neighbors know about it? Approve and show it on practice.

Answer:

In this lab, the static route on Router2:

```
[admin@Router2] > /ip route add dst-address=172.16.10.0/24 gateway=10.0.0.1
```

The *172.16.10.0/24* network is physically reachable on *Router2*. This network is not in OSPF and a static route tells *Router2* which gateway is reachable (*10.0.0.1*) [3].

Then redistribute is enabled on the OSPF instance:

```
[admin@Router2] > /routing ospf instance set [find name=default] redistribute=static
```

This setting tells OSPF to send the static routes as OSPF LSA type-5. In LSA type, Type-5 LSA = External route and Router2 becomes ASBR (Autonomous System Boundary Router).

5. Enable OSPF with authentication between the neighbors and verify it.

Answer:

OSPF authentication is already enabled everywhere in the lab. This authentication key is set the same on all four routers. This is the correct syntax for RouterOS v7.

Router 1 - OSPF (Area 0)

```
[admin@Router1] > /routing ospf instance add name=default router-id=1.1.1.1
```

Here, an OSPF instance has been created on Router1, named default, and its router-id is set to *1.1.1.1*. The OSPF router-id is the unique identity of the router, which is used in OSPF database, LSA, DR/BDR election [3].

```
[admin@Router1] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

This creates an area called area0 in OSPF with area-id = *0.0.0.0*.

Area 0 is the Backbone Area of OSPF, where all non-backbone areas are ultimately connected. Router1 is part of this area0 [3].

```
[admin@Router1] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Here *10.0.0.0/24* (Backbone LAN) is advertised in OSPF within area0.

Router1, Router2, Router3 are together in this network, so they detect each other as OSPF neighbors. OSPF authentication is enabled with *auth=simple auth-key=labpass* so all routers are given the same password [1], [2].

```
[admin@Router1] > /routing ospf interface-template add networks=192.168.1.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Router1's local LAN1 network *192.168.1.0/24* is also included in area0.

This allows other routers to learn about the *192.168.1.0/24* subnet via OSPF and get a route to Client1's network [3].

Router 2 - OSPF (Area 0)

```
[admin@Router2] > /routing ospf instance add name=default router-id=2.2.2.2
```

An OSPF instance is also created on Router2, with router-id = *2.2.2.2*. This is different from *Router1*'s router-id, so that there is no conflict in the OSPF database [3].

```
[admin@Router2] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

Router2 is also joined to backbone area0, just like *Router1*. This makes *Router2* part of the backbone and forms an OSPF adjacency with R1 and R3 [3].

```
[admin@Router2] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpass
```

Backbone network 10.0.0.0/24 is enabled in OSPF of Router2.

It appears as a neighbor with *Router2, Router1* and *Router3*. Here too, Authentication key labpass [3].

```
[admin@Router1] > /routing ospf interface-template add networks=192.168.1.0/24 area=area0 auth=simple auth-key=labpassh-key=labpass
```

Router2's local LAN2 (*192.168.2.0/24*) is advertised in area0 [1].

Router 3 - OSPF (Area 0 + Area 1)

Router3 is ABR

```
[admin@Router3] > /routing ospf instance add name=default router-id=3.3.3.3
```

An OSPF instance is created on *Router3* and the router-id is set to *3.3.3.3*. This router-id is used for LSAs and inter-area routing when acting as an ABR [3].

```
[admin@Router2] > /routing ospf area add name=area0 area-id=0.0.0.0 instance=default
```

Router3 is made part of backbone area0 so that it can participate in the backbone by becoming an OSPF neighbor with *Router1* and *Router2* [3].

```
[admin@Router3] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
```

Here a new OSPF area has been created - area1 (area-id *0.0.0.1*). This is the non-backbone area where *Router3* and *Router4* operate. *Router3* now stands on the boundary of area0 and area1 together, it is called ABR (Area Border Router).

Area 0

```
[admin@Router3] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0 auth=simple auth-key=labpass
```

Backbone link 10.0.0.0/24 is enabled in area0 for *Router3*. It creates OSPF adjacency with *Router1* and *Router2* and syncs the backbone LSDB.

Area 1

```
[admin@Router3] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1 auth=simple auth-key=labpass
```

Here, *Router3* and *Router4* have a point-to-point link (*10.0.1.0/30*) added to area1. This creates an OSPF neighbor relationship between R3 and R4, but this link is not a backbone, as it is part of Area1 [3].

```
[admin@Router3] > /routing ospf interface-template add networks=192.168.3.0/24 area=area1 auth=simple auth-key=labpass
```

Here, *Router3*'s LAN3 subnet (*192.168.3.0/24*) is advertised in area1.

The route of this subnet is propagated to both area1 and area0 via OSPF due to the presence of ABR [3].

Router 4 – OSPF (Area 1)

```
[admin@Router4] > /routing ospf instance add name=default router-id=4.4.4.4
```

An OSPF instance has been created on *Router4* with router-id *4.4.4.4*.

```
[admin@Router4] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
```

Router4 is only part of area1. It is not directly connected to backbone area0, but participates in Area1 through *Router3* [3].

```
[admin@Router4] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1 auth=simple auth-key=labpass
```

Router4's P2P link (*10.0.1.0/30*) is enabled in OSPF Area1 with *Router3*.

```
[admin@Router4] > /routing ospf interface-template add networks=192.168.4.0/24 area=area1 auth=simple auth-key=labpass
```

Router4's local *LAN4* subnet (*192.168.4.0/24*) is advertised in OSPF Area1. As a result, other routers R1, R2, R3 can learn about this network through OSPF and get a route to *Client4* [3].

Verification:

```
[admin@Router1] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.2 priority=128
      router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=5m43s timeout=37s

 1  D instance=default area=area0 address=10.0.0.3 priority=128
      router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=3m33s timeout=38s
```

This output shows two OSPF neighbors of Router1.

Neighbor: 10.0.0.2

Neighbor Router-ID = *2.2.2.2* (*Router2*) and State = Full, so adjacency is successful.

DR = *10.0.0.1* (*Router1*) and BDR = *10.0.0.2* (*Router2)*. Also, there is no Authentication mismatch.

Neighbor: 10.0.0.3

Neighbor Router-ID = *3.3.3.3* (*Router3*) and State = Full. DR / BDR are the same because there is election consistency in the backbone area. Timeout means after how many seconds does the adjacency break if no hello is received. This means that Authentication is working. As a result, Full adjacency is established on *Router1, Router2 & Router3* [3].

```
[admin@Router2] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=7m35s timeout=35s

 1  D instance=default area=area0 address=10.0.0.3 priority=128 router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=5m25s timeout=36s
```

The neighbor print of *Router2* shows two OSPF neighbors:

Neighbor (10.0.0.1 = Router1)

- Router-ID = *1.1.1.1*
- State = Full
- DR = *10.0.0.1*
- BDR = *10.0.0.2*

Neighbor (*10.0.0.3 = Router3*)

- Router-ID = *3.3.3.3*
- State = Full

Router2 has established a fully authenticated adjacency with both *Router1* and *Router3* on the backbone [2], [3].

```
[admin@Router3] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=6m46s timeout=34s

 1  D instance=default area=area0 address=10.0.0.2 priority=128 router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=6m46s timeout=34s

 2  D instance=default area=area1 address=10.0.1.2 priority=128 router-id=4.4.4.4 dr=10.0.1.1 bdr=10.0.1.2 state="Full" state-changes=6 adjacency=4m47s timeout=33s
```

Router3 has three neighbors:

Backbone Area0 neighbors

Neighbor: Router1 (10.0.0.1)

- RID = *1.1.1.1*
- State = Full
- Authentication is OK

Neighbor: Router2 (10.0.0.2)

- RID = *2.2.2.2*
- State = Full

Area1 neighbor

Neighbor: Router4 (10.0.1.2)

- RID = *4.4.4.4*
- State = Full
- This is Area1 adjacency
- If the authentication key mismatch between R3 and R4, the adjacency would never be Full.

Router3 authentication is working properly in both the backbone (Area0) and non-backbone (Area1) areas and this is the correct behavior of the ABR router [3].

```
[admin@Router4] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area1 address=10.0.1.1 priority=128 router-id=3.3.3.3 dr=10.0.1.1
      bdr=10.0.1.2 state="Full" state-changes=6 adjacency=5m52s timeout=38s
```

Router4 has only one neighbor (Router3), because it is only part of Area1.

- Neighbor Router-ID = **3.3.3.3** (Router3)
- State = **Full**
- DR = 10.0.1.1 (Router3)
- BDR = 10.0.1.2 (Router4 itself)
- Adjacency stable

Router4's authentication key matches Router3's, so they have created a Full adjacency. Here we see that neighbor state = Full.

6. Bonus: if one of the routers has multiple subnets, try to use route summarization.

Answer:

In this lab, *Router3* has multiple subnets in Area 1, such as *192.168.3.0/24* and *192.168.4.0/24.* Normally, these are propagated as separate LSAs to Area 0, which makes the OSPF database large and creates unnecessarily many entries in the routing table. To solve this problem, route summarization is used in OSPF. Since *Router3* is an ABR (Area Border Router) between Area 1 and Area 0, this command is applied only to *Router3*. These two /24 subnets in Area 1 are summarized together and declared as a larger subnet *192.168.3.0/23* [1].

```
[admin@Router3] > /routing ospf area range add area=area1 prefix=192.168.3.0/23 advertise=yes
```

In RouterOS v7, summarization is performed using the /routing ospf area range add command, which creates a summary prefix and allows it to be advertised [3]. Summarization thus makes the routing table smaller, cleaner, and more efficient, and only one consolidated route is visible in Area 0, even though Area 1 has two separate subnets. In this lab, summarization worked correctly and the summary prefix was displayed in the routing table on *Router1/Router2*, indicating that the ABR successfully implemented summarization.

TASK 3 – OSPF VERIFICATION

1. How can you check if you have a full adjacency with your router neighbor?

Answer:

To run OSPF, each router has to establish a permanent neighbor relationship with its neighboring routers. The highest and most stable state of this relationship is state = Full [1], [2]. Full adjacency

means that two routers have successfully completed the LSDB exchange and both accept each other as fully synchronized OSPF neighbors [3].

The following command is used on each router to verify OSPF adjacency [1]:

```
[admin@Router1] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.2 priority=128
      router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=1h16m7s timeout=34s

 1  D instance=default area=area0 address=10.0.0.3 priority=128
      router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=1h13m57s timeout=34s
[admin@Router1] >
```

Router1 has two backbone neighbors:

- Neighbor Router-ID *2.2.2.2* (*Router2*)
- Neighbor Router-ID *3.3.3.3* (*Router3*)

Both neighbors are in state= Full. This proves that *Router1* has formed a full adjacency with all its neighbors in Area0. Router1 has been elected as DR and *Router2* as BDR, which is a sign of a healthy OSPF election. This means that Adjacency is OK, no authentication or OSPF issues [3].

```
[admin@Router2] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=7m35s timeout=35s

 1  D instance=default area=area0 address=10.0.0.3 priority=128 router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=5m25s timeout=36s
```

Router2 also has two backbone neighbors, where:

- Neighbor Router-ID *1.1.1.1* (*Router1*)
- Neighbor Router-ID *3.3.3.3* (*Router3*)

Here both neighbors are showing state= Full. *Router2* shows *10.0.0.1* as DR and *10.0.0.2* as BDR according to a successful DR/BDR election in the backbone [3].

As a result, all adjacencies were FULL and OSPF LSDB exchange was completed.

```
[admin@Router3] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=6m46s timeout=34s

 1  D instance=default area=area0 address=10.0.0.2 priority=128 router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=6m46s timeout=34s

 2  D instance=default area=area1 address=10.0.1.2 priority=128 router-id=4.4.4.4 dr=10.0.1.1 bdr=10.0.1.2 state="Full" state-changes=6 adjacency=4m47s timeout=33s
```

Router3 is seen see two neighbors on the backbone and one neighbor on Area1:

Backbone (Area0) neighbors:

- Router1 - RID *1.1.1.1* = Full
- Router2 - RID *2.2.2.2* = Full

Area1 neighbor:

- Router4 - RID *4.4.4.4* = Full

Router3 is an ABR (Area Border Router), whose job is to forward LSAs between Area0 and Area1. All three neighbors are in the Full state, which proves that there is no adjacency problem from the ABR side (fully healthy) [2], [3].

```
[admin@Router4] > /routing ospf neighbor print
Flags: V - virtual; D - dynamic
 0  D instance=default area=area1 address=10.0.1.1 priority=128 router-id=3.3.3.3 dr=10.0.1.1
      bdr=10.0.1.2 state="Full" state-changes=6 adjacency=1h12m17s timeout=33s
[admin@Router4] >
```

Router4 is only in Area1, so it has only one neighbor Router3 - RID 3.3.3.3 = state= Full. This is also expected, because R3 to R4 is a point-to-point adjacency.

As a result, it is seen that Required adjacency OK, state=Full.

2. How can you check in the routing table which networks you received from neighbors?

Answer:

In OSPF, the networks that each router learns from its neighbors are added to the routing table as "Dynamic OSPF" routes. These routes can be viewed with the /ip route print where ospf command. A route entry with the "DAo" or "Do" flag indicates that it was learned from another router via an OSPF LSA [1], [2].

```
[admin@Router1] > /ip route print where ospf
Flags: D - DYNAMIC; I - INACTIVE, A - ACTIVE; o - OSPF; H - HW-OFFLOAD
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS     GATEWAY          DISTANCE
DAo  10.0.1.0/30     10.0.0.3%ether1       110
DIoH 172.16.10.0/24  10.0.0.1%ether1       110
DAo  192.168.2.0/23  10.0.0.3%ether1       110
DAo  192.168.2.0/24  10.0.0.2%ether1       110
DAo  192.168.4.0/24  10.0.0.3%ether1       110
[admin@Router1] >
```

The OSPF route table of Router1 shows that:

- *10.0.1.0/30* - came from the *Router3/R4* link
- *172.16.10.0/24* - static route redistributed by *Router2*
- *192.168.2.0/23* - summarization from Area1 (from Router3)
- *192.168.2.0/24* - *Router2*'s LAN
- *192.168.4.0/24* - *Router4*'s LAN

Router1 has correctly learned all remote LANs (R2, R3, R4) and redistributed static network neighbors [3].

```
[admin@Router2] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 10.0.1.0/30     10.0.0.3%ether1       110
DAo 192.168.1.0/24  10.0.0.1%ether1       110
DAo 192.168.2.0/23  10.0.0.3%ether1       110
DAo 192.168.4.0/24  10.0.0.3%ether1       110
[admin@Router2] >
```

From Router2 it can be seen that:

- *10.0.1.0/30 = Router3-R4* link
- *192.168.1.0/24 = Router1* LAN
- *192.168.2.0/23* = summarized network from *Router3*
- *192.168.4.0/24 = Router4* LAN

Router2 has learned the necessary routes from both the backbone and Area1 [3].

```
[admin@Router3] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 172.16.10.0/24  10.0.0.1%ether1       110
DAo 192.168.1.0/24  10.0.0.1%ether1       110
DAo 192.168.2.0/24  10.0.0.2%ether1       110
DAo 192.168.4.0/24  10.0.1.2%ether3       110
[admin@Router3] >
```

At Router3:

- ***172.16.10.0/24*** - *Router2*'s redistributed static route
- ***192.168.1.0/24*** - *Router1* LAN
- ***192.168.2.0/24*** - *Router2* LAN
- ***192.168.4.0/24*** - *Router4* LAN

As an ABR, *Router3* correctly received LSAs from both Backbone (Area0) and Area1 [3].

```
[admin@Router4] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 10.0.0.0/24     10.0.1.1%ether1       110
DAo 172.16.10.0/24  10.0.1.1%ether1       110
DAo 192.168.1.0/24  10.0.1.1%ether1       110
DAo 192.168.2.0/24  10.0.1.1%ether1       110
DAo 192.168.3.0/24  10.0.1.1%ether1       110
[admin@Router4] >
```

At Router4:

- *10.0.0.0/2* - Backbone LAN
- *172.16.10.0/24* - redistributed static route
- *192.168.1.0/24* - Router1 LAN
- *192.168.2.0/24* and *192.168.3.0/24* - Area0 & Area1 networks

Router4 has learned all the subnets of the entire OSPF network through its Area1 neighbor (*Router3*) [2], [3].

3. Use traceroute to verify that you have a full OSPF network.

Answer:

When OSPF is working properly, the traceroute command shows all the intermediate routers as one "hop" after another. Traceroute actually shows which routers the packet passed through on its way to its destination by incrementing the TTL at each hop. It is therefore a very important tool for end-to-end routing verification [2], [6].

VPCS Client1:

```
PC1> trace 192.168.1.10
traceroute to 192.168.1.10, 8 hops max
 1 192.168.1.10     0.001 ms
```

Client1 is tracerouted to its own IP, so only 1 hop is shown - itself (*192.168.1.10*). This indicates that the local stack is OK, but it is more important to traceroute to a client on another subnet to verify the OSPF path [6].

VPCS Client4:

```
PC4> trace 192.168.1.10
trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop
 1   192.168.4.1   0.793 ms  0.693 ms  0.560 ms
 2   10.0.1.1   1.212 ms  1.268 ms  1.623 ms
 3   10.0.0.1   1.897 ms  2.066 ms  2.009 ms
 4   *192.168.1.10   3.354 ms (ICMP type:3, code:3, Destination port unreachable)
```

It can be seen that in traceroute, each intermediate router is shown as a hop in exactly this way.

ICMP type 3 - Destination port unreachable is shown because:

In VPCS, trace usually sends probes using UDP high port. No service is listening on that UDP port on *Client1*, so the end host sends ICMP type 3, code 3 (Destination port unreachable). This is completely normal behavior. The important thing is that the host is reachable, so OSPF or IP routing is working properly [2], [6].

4. Which router is selected as DR and which one is BDR?

Answer:

In OSPF, DR (Designated Router) and BDR are elected only in multi-access networks. In the topology, the multi-access network is only Backbone LAN: *10.0.0.0/24*, where three routers are connected:

- *Router1 (10.0.0.1)*
- *Router2 (10.0.0.2)*
- *Router3 (10.0.0.3)*

Router4 is on a point-to-point link (10.0.1.0/30) so DR/BDR is not elected here [3].

Router1, Router2, Router3

```
[admin@Router1] > /routing ospf neighbor print detail
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.2 priority=128
      router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=1h26m51s timeout=40s

 1  D instance=default area=area0 address=10.0.0.3 priority=128
      router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6
      adjacency=1h24m41s timeout=40s
```

At Router1:

Neighbor 1:

- router-id = *2.2.2.2* - *Router2*
- dr = *10.0.0.2* - *Router2* is DR
- bdr = *10.0.0.2*

Neighbor 2:

- router-id = *3.3.3.3* - *Router3*
- dr = *10.0.0.1*

But it doesn't appear in two places. This is because even though R1 is DROTHER, the LSAs context is different. The actual decision comes from the screenshots of *Router2* and *Router3*.

```
[admin@Router2] > /routing ospf neighbor print detail
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=1h26m55s timeout=36s

 1  D instance=default area=area0 address=10.0.0.3 priority=128 router-id=3.3.3.3 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=1h24m45s timeout=36s
[admin@Router2] >
```

At Router2:

Neighbor 1:

- router-id = *1.1.1.1 - Router1*
- dr = *10.0.0.1*
- bdr = *10.0.0.2*

Neighbor 2:

- router-id = *3.3.3.3*
- dr = *10.0.0.1*
- bdr = *10.0.0.2*

```
[admin@Router3] > /routing ospf neighbor print detail
Flags: V - virtual; D - dynamic
 0  D instance=default area=area0 address=10.0.0.1 priority=128 router-id=1.1.1.1 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=1h24m48s timeout=33s

 1  D instance=default area=area0 address=10.0.0.2 priority=128 router-id=2.2.2.2 dr=10.0.0.1 bdr=10.0.0.2 state="Full" state-changes=6 adjacency=1h24m48s timeout=33s

 2  D instance=default area=area1 address=10.0.1.2 priority=128 router-id=4.4.4.4 dr=10.0.1.1 bdr=10.0.1.2 state="Full" state-changes=6 adjacency=1h22m49s timeout=32s
[admin@Router3] >
```

At Router3:

Neighbor 1 - Router1:

- dr = *10.0.0.1*
- bdr = *10.0.0.2*

Neighbor 2 - Router2:

- dr = *10.0.0.1*
- bdr = *10.0.0.2*

Based on the neighbor detail output from *Router1*, *Router2*, and *Router3*, *Router1* (*10.0.0.1*) was elected as the DR, and *Router2 (10.0.0.2*) was elected as the BDR. *Router3* operates as a DROTHER [3].

5. Check what is the cost for each network that has been received by OSPF in the routing table.

Answer:

The OSPF cost metric tells which path is more "cheap" / efficient. Here Lower cost = Better path and OSPF cost is derived from interface cost.

Initially there was default cost, later *cost=5* was set on backbone (*10.0.0.0/24*) [3].

```
[admin@Router1] > /ip route print where ospf
Flags: D - DYNAMIC; I - INACTIVE, A - ACTIVE; o - OSPF; H - HW-OFFLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS     GATEWAY          DISTANCE
DAo  10.0.1.0/30     10.0.0.3%ether1       110
DIoH 172.16.10.0/24  10.0.0.1%ether1       110
DAo  192.168.2.0/23  10.0.0.3%ether1       110
DAo  192.168.2.0/24  10.0.0.2%ether1       110
DAo  192.168.4.0/24  10.0.0.3%ether1       110
[admin@Router1] >
```

At Router1:

- All routes DAo / DIoH - Dynamic Active OSPF routes
- Distance = 110 = OSPF default AD

Cost is not shown directly in the table, so the metric is seen from the OSPF route table and the Gateway shows which neighbor *Router1* learned this network from [3].

```
[admin@Router2] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 10.0.1.0/30     10.0.0.3%ether1       110
DAo 192.168.1.0/24  10.0.0.1%ether1       110
DAo 192.168.2.0/23  10.0.0.3%ether1       110
DAo 192.168.4.0/24  10.0.0.3%ether1       110
[admin@Router2] >
```

At Router2:

- All routes OSPF-learned
- Default distance is shown

Cost override command succeeded, but routing table distance does not show cost.

```
[admin@Router3] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 172.16.10.0/24  10.0.0.1%ether1       110
DAo 192.168.1.0/24  10.0.0.1%ether1       110
DAo 192.168.2.0/24  10.0.0.2%ether1       110
DAo 192.168.4.0/24  10.0.1.2%ether3       110
[admin@Router3] >
```

At Router3:

Router3 is getting mix routes from backbone (Area0) and Area1 and both summarized + not summarized routes are visible. All OSPF path distance = 110.

```
[admin@Router4] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 10.0.0.0/24     10.0.1.1%ether1       110
DAo 172.16.10.0/24  10.0.1.1%ether1       110
DAo 192.168.1.0/24  10.0.1.1%ether1       110
DAo 192.168.2.0/24  10.0.1.1%ether1       110
DAo 192.168.3.0/24  10.0.1.1%ether1       110
[admin@Router4] >
```

At Router4:

Router4 learned all routes from *Router3* and all OSPF-learned. Cost = default path metric and Summarized *192.168.3.0/23* network-also came up correctly.

interface-template ৻ cost override:

```
[admin@Router1] > /routing ospf interface-template set [find networks=10.0.0.0/24] cost=5
[admin@Router1] >
```

```
[admin@Router2] > /routing ospf interface-template set [find networks=10.0.0.0/24] cost=5
[admin@Router2] >
```

```
[admin@Router3] > /routing ospf interface-template set [find networks=10.0.0.0/24] cost=5
[admin@Router3] >
```

```
[admin@Router4] > /routing ospf interface-template set [find networks=10.0.0.0/24] cost=5
[admin@Router4] >
```

I modified the interface-template cost for the backbone network (*10.0.0.0/24*) to *cost=5* on all routers. This confirms that the OSPF cost value for that link is now overridden and will be used for SPF calculations. All routers successfully installed their OSPF-learned networks with the correct cost configuration [1], [3].

BONUS - Multi-Area OSPF Network

1. In the case if until now every router was in area 0, try to create more networks and assign them to different OSPF areas.

Answer:

Initially all routers were in OSPF Area 0. We converted the design into a multi-area OSPF topology by keeping *Router1* and *Router2* as internal routers in Area 0 and making *Router3* an ABR, connecting Area 0 and Area 1 [1], [2].

Area 0 (Backbone)

- Router1, Router2, Router3
- Network: *10.0.0.0/24*

Area 1 (New Area)

- Link R3-R4: *10.0.1.0/30*
- R3 LAN (*LAN3*): *192.168.3.0/24*
- R4 LAN (*LAN4*): *192.168.4.0/24*

Router roles:

- *Router1*: Internal router (Area0 only)
- *Router2*: Internal router (Area0 only)
- *Router3*: ABR (Area0 + Area1)
- *Router4*: Internal router (Area1 only)

Router3 - Creating Area1 + Assign networks

Router3 is the **ABR**, so Area1 must be created here.

```
[admin@Router3] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
```

Add networks to Area1:

```
[admin@Router3] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1
[admin@Router3] > /routing ospf interface-template add networks=192.168.3.0/24 area=area1
```

Here a new OSPF Area named area1 has been created - 0.0.0.1. The R3-R4 p2p link *10.0.1.0/30* is in Area1 and R3's LAN *192.168.3.0/24* is in Area1. The same router is again a member of backbone *10.0.0.0/24* in Area0. As a result, *Router3* is now participating in both Area0 and Area1, which is ABR [3].

Router4 - Belongs completely to Area1

```
[admin@Router4] > /routing ospf area add name=area1 area-id=0.0.0.1 instance=default
[admin@Router4] > /routing ospf interface-template add networks=10.0.1.0/30 area=area1
[admin@Router4] > /routing ospf interface-template add networks=192.168.4.0/24 area=area1
```

The same area1 has been created on R4 (same area-id = 0.0.0.1). Both R3-R4 link + R4 LAN are in Area1. R4 is no longer a member of any subnet in Area0 where Area1 is an internal router [1].

Router1 & Router2

```
[admin@Router1] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0
[admin@Router3] > /routing ospf interface-template add networks=192.168.3.0/24 area=area1
```

Router2:

```
[admin@Router2] > /routing ospf interface-template add networks=10.0.0.0/24 area=area0
[admin@Router2] > /routing ospf interface-template add networks=192.168.2.0/24 area=area0
```

Backbone *10.0.0.0/24* + *Router1* LAN *192.168.1.0/24* are both in Area0 and the Internal router of Area0 has no Area1. Backbone + R2 LAN are both in Area0 and have no part of Area1 which is basically a pure Area0 router [3].

2. Why every area has to be connected to area 0?

Answer:

Area 0 is the Backbone Area which is the center of OSPF. In OSPF, each non-backbone Area (Area 1, Area 2, Area 3..) first sends its LSAs to Area 0, then Area 0 distributes them to other Areas. OSPF is a hierarchical protocol where Area-to-Area communication follows a rigid tree structure, whose root is Area 0. Area 1 and Area 2 cannot exchange routes directly, so all their routing information has to go through Area 0. 3. All non-backbone Areas must be connected to the Backbone. Because, summary LSA, inter-area LSA, and default route exchange of each Area is centered on Area 0. 4. If no Area is connected to Area0 - OSPF cannot provide loop-free stable routing. LSA propagation breaks and Inter-area routes become unreachable. This is why OSPF backbone follows a mandatory loop-free topology structure [1], [3].

3. What can you do if you have an area X which is not connected to area 0, but to area Y?

Answer:

A logical/virtual connection has been created with the virtual link backbone.

- Area1 is the transit area
- Area2 is the disconnected area (AreaX)
- A virtual link will be required between ABR of Area1 to ABR of Area2

On ABR1:

```
[admin@Router4] > /routing ospf virtual-link add transit-area=area1 neighbor-id=<ABR1-Router-ID>
bad command name virtual-link (line 1 column 15)
```

On ABR2:

```
[admin@Router2] > /routing ospf virtual-link add transit-area=area1 neighbor-id=<ABR1-Router-ID>
bad command name virtual-link (line 1 column 15)
```

This virtual link logically connects AreaX to Area0. But MikroTik RouterOS v7 does not support OSPF virtual links, so I cannot configure a virtual-link command. The only solution in RouterOS v7 is to redesign the topology so that every non-backbone area (Area X) is physically or logically connected to Area 0 [1], [3].

4. Verify that Area X can reach all networks.

Answer:

Ping tests

Router4 to Router1 LAN:

```
[admin@Router4] > ping 192.168.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.1.1                               56  63 1ms459us
    1 192.168.1.1                               56  63 4ms198us
    2 192.168.1.1                               56  63 3ms867us
    3 192.168.1.1                               56  63 3ms768us
    4 192.168.1.1                               56  63 3ms435us
    5 192.168.1.1                               56  63 3ms645us
    6 192.168.1.1                               56  63 3ms154us
    sent=7 received=7 packet-loss=0% min-rtt=1ms459us avg-rtt=3ms360us max-rtt=4ms198us
```

7 packets were received and packet-loss=0%. *Router4* in Area1 was able to reach Router1 LAN (*192.168.1.1*) in Area0 directly. This means Area1 to Area0 connectivity was OK.

Router1 to Router4 LAN:

```
[admin@Router1] > ping 192.168.4.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.4.1                               56  63 1ms544us
    1 192.168.4.1                               56  63 3ms453us
    2 192.168.4.1                               56  63 3ms494us
    3 192.168.4.1                               56  63 3ms468us
    4 192.168.4.1                               56  63 3ms975us
    5 192.168.4.1                               56  63 3ms448us
    sent=6 received=6 packet-loss=0% min-rtt=1ms544us avg-rtt=3ms230us
   max-rtt=3ms975us
```

Reply was received and packet-loss=0%

Router1 in Area0 to Router4 in Area1 pinged on LAN. This means there is reachability in both directions.

Traceroute tests

From *Client4* (LAN4, Area1) to *Client1* (LAN1, Area0):

```
Clint4> trace 192.168.1.10
trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop
 1   192.168.4.1   1.119 ms  1.170 ms  1.018 ms
 2   10.0.1.1   2.512 ms  2.537 ms  2.607 ms
 3   10.0.0.1   4.014 ms  3.491 ms  3.908 ms
 4   *192.168.1.10   5.349 ms (ICMP type:3, code:3, Destination port unreachable)

Clint4>
```

Hop 1-3 shows the exact path that the network design expected. The last line shows ICMP type:3, code:3, meaning that the high UDP port that traceroute sent the probe to is not open on Client1, so it shows Destination port unreachable. This proves that the traceroute reached the correct host (192.168.1.10), and then the host replied that the port is not available [3].

Routing Table check

At Router1:

```
[admin@Router1] > /ip route print where ospf
Flags: D - DYNAMIC; I - INACTIVE, A - ACTIVE; o - OSPF; H - HW-OFFLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
     DST-ADDRESS     GATEWAY          DISTANCE
DAo  10.0.1.0/30     10.0.0.3%ether1       110
DIoH 172.16.10.0/24  10.0.0.1%ether1       110
DAo  192.168.2.0/23  10.0.0.3%ether1       110
DAo  192.168.2.0/24  10.0.0.2%ether1       110
DAo  192.168.4.0/24  10.0.0.3%ether1       110
```

Router1 learned Area1's network *192.168.4.0/24* and p2p *10.0.1.0/30* from its OSPF neighbor (R3).

Area1 to Area0 advertisement OK [1].

At Router4:

```
[admin@Router4] > /ip route print where ospf
Flags: D - DYNAMIC; A - ACTIVE; o - OSPF
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY          DISTANCE
DAo 10.0.0.0/24     10.0.1.1%ether1       110
DAo 172.16.10.0/24  10.0.1.1%ether1       110
DAo 192.168.1.0/24  10.0.1.1%ether1       110
DAo 192.168.2.0/24  10.0.1.1%ether1       110
DAo 192.168.3.0/24  10.0.1.1%ether1       110
```

Router4 shows that *Router4* of Area1 has learned all subnets of the entire network (Area0 + other Area1 LAN + external static) via OSPF. This confirms that Area X (Area1) can reach all networks. Therefore, it is proven that Area X (Area1) is fully reachable with the entire OSPF network [1], [3].

References

[1] MikroTik, “RouterOS Documentation,” *MikroTik Docs*. Available: https://help.mikrotik.com/docs/display/ROS/RouterOS

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[4] GNS3, “VPCS - Virtual PC Simulator,” *GNS3 Docs*. Available: https://docs.gns3.com/docs/vpcs/

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[6] M. Kerrisk, “traceroute(8) - Linux manual page,” *man7.org Linux man-pages*. Available: https://man7.org/linux/man-pages/man8/traceroute.8.html